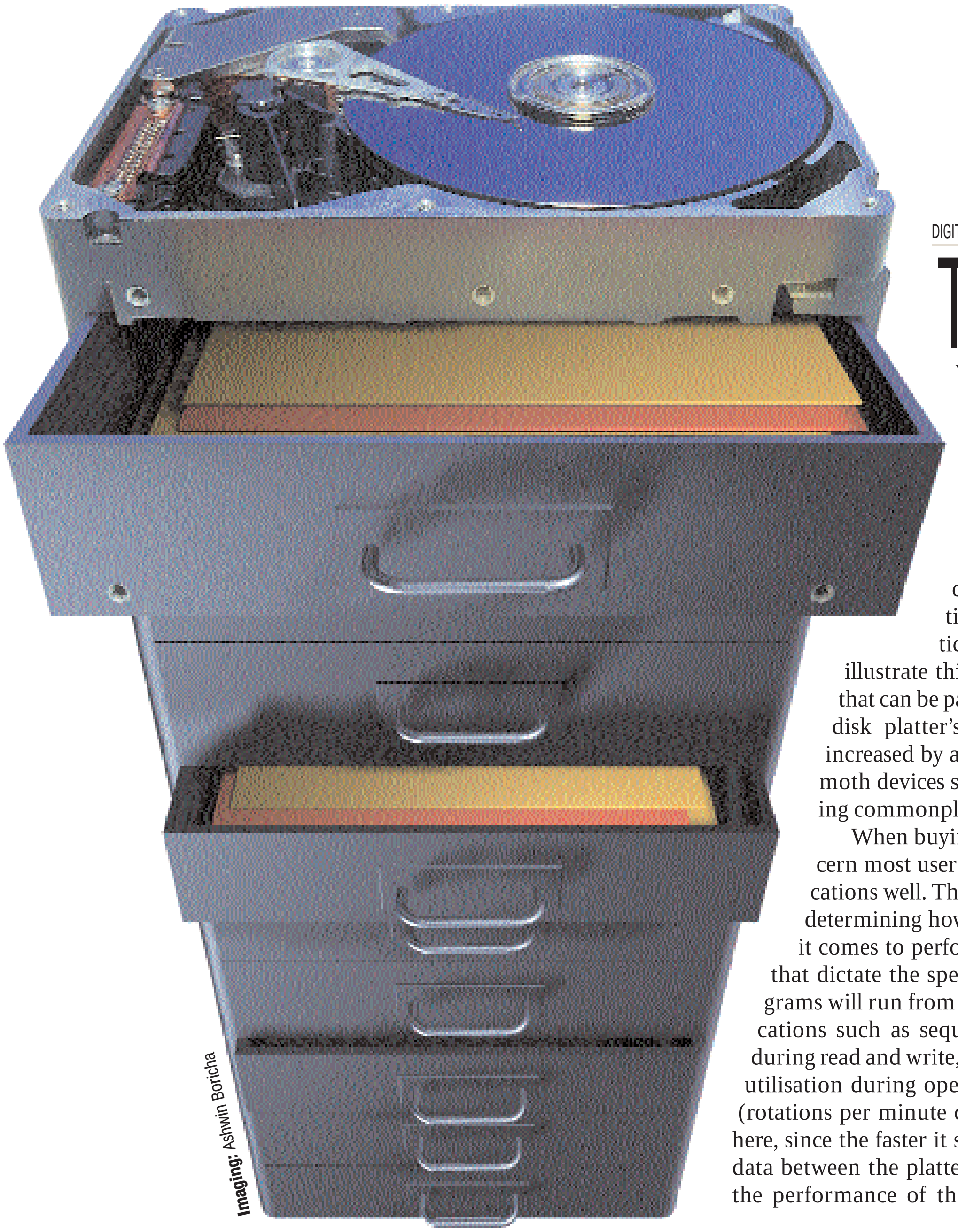
IN TEST: 15 IDE HARD DISK DRIVES

With hard disks like these, you can give your PC all the space it needs!

Room for more



Imaging: Ashwin Boricha

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The storehouse of your computer, hard disks are marvels of modern engineering that glean the best from fields such as mechanical, electronic, aeronautic as well as chemical engineering. The construction of a hard disk involves very precise and micron-level tolerances, which is now possible with the help of highly sophisticated machinery. Right from the time the first 20 MB hard disk made its appearance to the 100 GB hard drives available today, the increase in capacity of these is a very accurate indication of the progress of computing in particular and technology in general. To illustrate this, the areal density (the amount of data that can be packed into a single unit of area on the hard disk platter's surface) of the hard disk drive has increased by around 5 million times. Therefore, mammoth devices such as the 100 GB drives are fast becoming commonplace nowadays.

When buying a hard disk, the first and primary concern most users have is whether it will serve their applications well. Then there's also cost that plays a role when determining how large a hard disk you can afford. When it comes to performance, there are a range of parameters that dictate the speed at which your applications and programs will run from your drive. These parameters are specifications such as sequential and random data transfer rates during read and write, access time of the hard disk and the CPU utilisation during operation. The spindle speed of your drive (rotations per minute or rpm) will also play an important role here, since the faster it spins, the faster will it be able to transfer data between the platter and the read/write heads. To evaluate the performance of these parameters, we used a selection of